Online Case Study: Portfolio Management Simulation

BUS 104: Decision Analysis and Management Science

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UC RIVERSIDE | UNIVERSITY OF CALIFORNIA | School of Business
A. GARY ANDERSON GRADUATE SCHOOL OF MANAGEMENT

Case Study Overview: From 3 Stocks to 25 Global Companies

Putting Nonlinear Programming Into Action (10% of Course Grade)

In our Nonlinear Programming lectures, you learned how to find optimal portfolio weights w_i that balance **expected return** and **portfolio variance (σ_p^2)** using `scipy.optimize`. Now you will apply those tools in a live multiplayer simulation!

What You Will Do in the Simulation

- **Platform:** Harvard Business Publishing / Finsimco.
- **Universe:** Allocate capital across the **25 largest global companies** over **4 quarters**.
- **Teams: 3 to 5 students** managing a **Hedge Fund** or **Pension Fund**.
- **Time Window:** ~ 2.5–3 hours total across **1 week (Nov 18 – Nov 25)**.

Key Rules at a Glance

- **Purchase Cutoff: Tuesday, Nov 17 at 11:59 PM** (teams formed Wed, Nov 18).
- **Guaranteed 80% Floor:** Every active student completing a valid attempt earns $\geq 80\%$.
- **Free-Rider Protection:** Individual Canvas upload + peer sign-off required.

Step 1: Course Pack Purchase & Hard Cutoff Deadline

Hard Purchase Deadline: Tuesday, November 17 at 11:59 PM

- **Harvard Course Pack Link:** `https://hbsp.harvard.edu/import/1456741`
- Because the simulation **only operates in teams**, teams cannot be created until the roster of purchased accounts is locked.
- **Pay-to-Play Roster Lock:** Only students registered in the Harvard Course Pack by **Tuesday, Nov 17 at 11:59 PM** will be placed on a team on Wednesday, Nov 18. **If you miss this cutoff, you cannot join a team later and will receive 0%.**

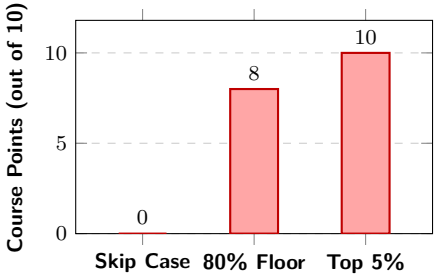
CRITICAL: Exact Canvas Email & Name Match Required (From Syllabus)

- When registering on Harvard Business Publishing, you **must use the exact same email address and full name** as your UCR Canvas (`eLearn.ucr.edu`) account.
- **Per the BUS 104 Syllabus:** *"If the email is different, or if your name under which you purchase the course pack is different, you will not receive credit for your case work."*

Why You Should Not Skip This Case: Look at the Grade Math!

How the 10% Weight Impacts Your Grade

- **Syllabus Floor (Table 1):** Any valid attempt is guaranteed **at least 80%** ($8.0/10$ course points).
- **Skip the case:** You get **0 / 10 points**—an automatic **full letter-grade drop** in BUS 104 (e.g., A- $\rightarrow$ B-, or B $\rightarrow$ C).
- **Participate & submit:** You lock in **8.0 to 10.0 points**.
- **1st vs. Last Place Among Active Teams:** The gap between 1st place (100%) and bottom quartile (80%) is **at most 2.0% of your total course grade**!



Participating locks in $+8.0$ pts; rank adds 0 to $+2.0$ pts!

Team Formation & Your Two Client Mandates

How Teams Are Formed (3 to 5 Students per Team)

- **Roster Lock:** Teams are created on **Wednesday, Nov 18** exclusively from students registered in the Harvard Course Pack by the **Tuesday, Nov 17 (11:59 PM)** cutoff.
- **Optional Teammate Preference:** To work with 1–2 specific classmates, ensure **all of you** have bought the pack and submit their names on Canvas by **Tue, Nov 17 at 11:59 PM**.

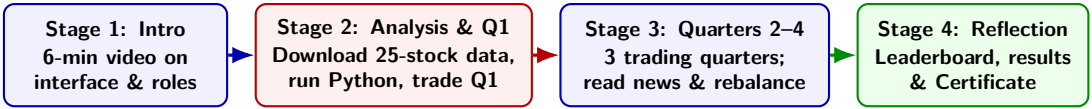
Odd Teams (1, 3, 5, . . .): Hedge Fund

- **Client:** High-growth Hedge Fund.
- **Goal:** Beat other **Hedge Fund teams** by maximizing your portfolio **Sharpe Ratio**.
- **Tools:** May borrow funds (leverage), concentrate or diversify, or hold cash.

Even Teams (2, 4, 6, . . .): Pension Fund

- **Client:** Conservative Pension Fund.
- **Goal:** Beat other **Pension Fund teams** by maximizing your **Sharpe Ratio subject to**:
- **Volatility Cap:** Portfolio volatility (σ_p) must stay **below the Pension Fund median**!

Inside the Simulation: The 4-Stage Gameflow ($\sim$ 2.5–3 Hours Total)



Stage 2 (Analysis & Quarter 1)

- Read your client mandate in **Mission** & **Guidance**.
- Download the historical prices for the 25 global stocks (Apple, Microsoft, NVIDIA, etc.).
- Optimize Q1 weights and submit trades.

Stage 3 (Quarters 2, 3 & 4)

- Each quarter brings realistic **newsfeeds, newsletters, and alerts** (e.g., *OPEC Cuts Oil Production*).
- Evaluate how news impacts expected returns/risks and **rebalance** your portfolio!

1-Week Case Study Schedule (Completed Before Thanksgiving)

1-Week Window: Wednesday, Nov 18 – Wednesday, Nov 25

Finishing by **Wednesday, Nov 25 at 11:59 PM** ensures nobody has to coordinate group meetings over Thanksgiving break!

Date & Time	Milestone	Required Action
Tue, Nov 17 (11:59 PM)	**HARD PURCHASE CUTOFF**	Buy course pack at `hbsp.harvard.edu/import/1456741` using your **exact Canvas email & name**.
Wed, Nov 18 (After Class)	**Teams Assigned & Game Opens**	Check Canvas for your Team # (Odd = Hedge Fund, Even = Pension Fund). **Intro + Analysis & Q1** opens.
Thu, Nov 19 – Sun, Nov 22	**Stage 2: Analysis & Quarter 1**	Meet with your team, run Python optimization on the 25-stock data, and submit **Quarter 1 trades**.
Mon, Nov 23 – Tue, Nov 24	**Stage 3: Quarters 2, 3 & 4**	Review market news alerts and execute your team's **Quarter 2, 3, and 4 rebalancing trades**.
Wed, Nov 25 (11:59 PM)	**FINAL SUBMISSION DUE**	Download your **Certificate of Completion** from Stage 4 and upload your **Individual Canvas Deliverable**.

Scaling Our Lecture Python Code from 3 Stocks to 25 Stocks

Instead of typing 25 variances and 300 covariances by hand, let `pandas` and `numpy` compute `mu` and `Sigma` directly from the downloaded Excel file in **Google Colab**:

```
import pandas as pd, numpy as np
from scipy.optimize import minimize, LinearConstraint

prices = pd.read_excel("historical_data.xlsx", index_col=0)
returns = prices.pct_change().dropna()
mu = returns.mean().values * 252          # Annualized expected returns (25 stocks)
Sigma = returns.cov().values * 252        # Annualized 25x25 covariance matrix

def neg_sharpe(w, rf=0.03):               # Minimize NEGATIVE Sharpe Ratio
    return -(np.dot(mu, w) - rf) / np.sqrt(np.dot(w, np.dot(Sigma, w)))

n = len(mu)
res = minimize(neg_sharpe, x0=np.ones(n)/n, bounds=[(0, 1)]*n,
               constraints=[LinearConstraint(np.ones(n), 1.0, 1.0)])
```

Tip for Even Teams (Pension Funds): You can also minimize portfolio variance $\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$ subject to a target return constraint to keep your volatility safely below the Pension Fund median!

Grading Policy: Relative Performance & The 80% Floor

How Your Rank Is Computed

1. **Fair Mandate Pools:** Teams are ranked by **Sharpe Ratio** *within their own mandate pool* (Hedge Funds vs. Hedge Funds; Pension Funds vs. Pension Funds).
2. **Natural Tie Handling:** Identical scores receive the exact same percentile rank and grade.
3. **80% Guaranteed Floor:** No score below **80%** is assigned for valid submitted work!

Table 1: Syllabus Scoring Scale (10% of Grade)

Percentile	Case Score	Class Position
95–100	**100%**	Top $\sim 5\%$
90–94.9	**95%**	Next ~ 5–10%
85–89.9	**91%**	Upper tier
80–84.9	**88%**	Upper-middle
75–79.9	**86%**	Middle to upper-middle
>0–74.9	**80%**	**Guaranteed Floor**
No Attempt	**0% (F)**	Non-submission / Ghost

Required Deliverables (Due Wednesday, Nov 25 at 11:59 PM)

Team Gameplay in Finsimco + Individual Canvas Check-In

Even though your team trades a shared portfolio inside Finsimco, **every student must submit an individual deliverable packet on Canvas** by **Wednesday, Nov 25 at 11:59 PM** to claim their team's score:

1. **Your Personal Finsimco Certificate of Completion (PDF or Screenshot):**
   - In Stage 4 (*Reflection*), click **Certificates and Feedback** on the left menu bar to download your personal certificate showing your name and completion.
2. **Your Team's Optimization File (`.ipynb` Notebook or Spreadsheet) + 1-Paragraph Summary:**
   - Upload the Google Colab notebook (`.ipynb`) or model used for Quarter 1, plus 1 short paragraph explaining how your team rebalanced in Quarters 2–4.
3. **Mandatory 1-Line Peer Contribution Sign-Off (in Canvas Submission Comments):**
   - Paste and complete in the Canvas comment box:
     `"All teammates contributed fairly: [Yes / No] (If No, state who did not participate)."`

FAQ: "What If My Teammates Don't Work and Leave It All to Me?"

How We Protect Hard-Working Students from Free-Riders

We enforce two strict safeguards so no student can coast on a teammate's work—and no hard-working student gets hurt by a flaky group:

1. Buying the Case $\neq$ Automatic 80%

- Simply buying the course pack to get placed on a team **does not earn credit**.
- The Finsimco **Admin Panel (*Live Monitoring*)** tracks every student's logins and stage activity.
- If teammates report on their Canvas Peer Sign-Off that you ghosted—and the Admin Panel confirms no participation or you fail to submit on Canvas—you receive **0% ("No Attempt")**.

2. Solo-Carry Protection for Active Members

- **Your grade cannot be tanked by ghosting teammates:** Any valid completion earns at least the **80% syllabus floor** ($8/10$ course points).
- **Protected Floor Bump:** If your teammates ghost and you are documented carrying the portfolio solo (or as a stranded pair), **your minimum floor score is bumped to** $\geq 88\%$, while non-participating teammates receive **0%**.

Your 4-Step Action Checklist

What You Need to Do Starting Today

1. **Step 1 — Buy the Course Pack by Tuesday, Nov 17 at 11:59 PM:**
   - Go to `https://hbsp.harvard.edu/import/1456741` and register with your **exact UCR Canvas email address and full name**.
2. **Step 2 — (Optional) Submit Teammate Preferences by Tuesday, Nov 17 at 11:59 PM:**
   - If you want to work with 1–2 specific classmates, verify that **all of you** have bought the course pack first and submit your request on Canvas before the cutoff.
3. **Step 3 — Check Your Team Assignment on Wednesday, Nov 18:**
   - Meet with your team (Thu Nov 19 – Sun Nov 22) to run your Python optimization and place Quarter 1 trades, then complete Quarters 2–4 on Mon Nov 23 – Tue Nov 24.
4. **Step 4 — Upload Your Individual Canvas Deliverable by Wednesday, Nov 25 at 11:59 PM:**
   - Upload your (1) Finsimco Certificate, (2) Python `.ipynb` / model + 1-paragraph summary, and (3) 1-line Peer Contribution Sign-Off before Thanksgiving break!